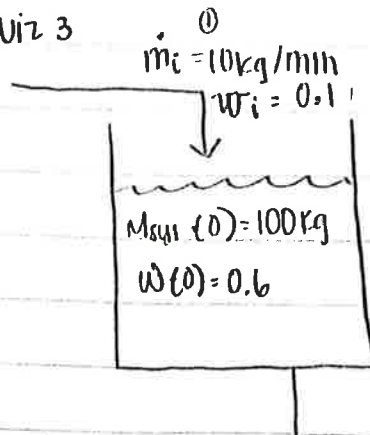


Kotwis 4

Patrycja Kotwis

Quiz 3



$$\text{total mass} = \frac{dm_{\text{sys}}}{dt} = \dot{m}_i - \dot{m}_e$$

species balance A: constant $w_e = w$

$$A: \frac{dm_{\text{sys}}}{dt} = \dot{m}_{Ai} - \dot{m}_{Ae}$$

$$m_{\text{sys}} = \int_{t_0}^{t_f} \dot{m} dt$$

$$\dot{m}_{Ai} = w_{Ai} \dot{m}_i$$

$$\dot{m}_{Ae} = w_{Ae} \dot{m}_e$$

$$m_{\text{sys}} = w_A \cdot m_{\text{sys}}$$

$$\frac{d(w_A m_{\text{sys}})}{dt} = w_{Ai} \dot{m}_i - w_{Ae} \dot{m}_e$$

good

$$\frac{dw_A}{dt} (m_{\text{sys}}) = (w_{Ai} - w_{Ae})(\dot{m}_i - \dot{m}_e)$$

$$\int_{w_0}^{w_f} \frac{1}{(w_{Ai} - w_{Ae})} dw_A = \int_{t_0}^{t_f} \frac{\dot{m}}{m_{\text{sys}}} dt = -\ln \left| \frac{w_{Ai} - w_{Ae}}{w_{Ai} - w_{Ae,0}} \right| = \frac{\dot{m}_{\text{sys}}}{m_{\text{sys}}} (t_f - t_0)$$

$$-\ln |w_{A\text{sys}}| =$$

$$t_f = \frac{m_{\text{sys}}}{\dot{m}} \left| \frac{w_{Ai}}{w_{Ai} - w_{Ae}} \right|$$

$$m_{\text{sys}} = \frac{w_A}{m_{A\text{sys}}} = \frac{w_A}{m_{A\text{sys}}} e^{-(t+4.6)}$$

$$= \frac{w_A}{?}$$

interesting

stays in $\frac{d}{dt}$ use product rule

Parvya Kaur

a)

$$\frac{dm_{sys}}{dt} = m_i - m_e$$

total mass

10-15

$$\int_{m_{sys}(0)}^{m_{sys}(t)} \frac{1}{m_i - m_e} dm_{sys} = \int dt$$

$$\frac{dm_{sys}}{dt} = -5$$

$$-\ln |m_i - m_e| \Big|_{m_0}^{m_f} = t_f - t_0$$

integration is wrong

$$-\ln |m_i - m_e| + C = t$$

$$-\ln |m_{sys}| + C = t$$

$$-\ln |100| + C = 0 \quad \leftarrow @ t=0$$

$$C = 4.6 \quad \rightarrow -\ln |m_{sys}| + 4.6 = t$$

$$-\ln |m_{sys}| = t - 4.6$$

$$\ln |m_{sys}| = -t + 4.6$$

$$m_{sys} = e^{(-t + 4.6)}$$

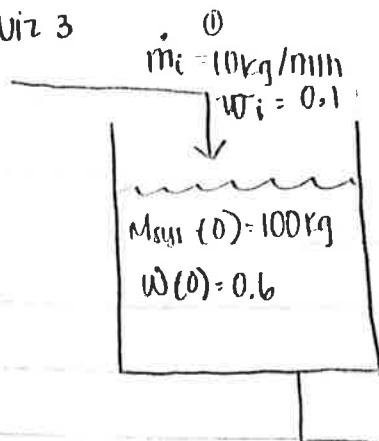
@ t=0 min

$$m_{sys} = e^{(-10 + 4.6)}$$

$$m_{sys} = 0.0045 \text{ kg}$$

Quiz 3

patrycja kotwiz



$$\text{total mass} = \frac{dm_{\text{sys}}}{dt} = \dot{m}_i - \dot{m}_e$$

species balance A:
constant

$$A: \frac{dm_{\text{sys}}}{dt} = \dot{m}_{Ai} - \dot{m}_{Ae}$$

$$m_{\text{sys}} = \int_{t_0}^{t_f} \dot{m} dt$$

$$\dot{m}_{Ai} = w_{Ai} \dot{m}_i$$

$$\dot{m}_{Ae} = w_{Ae} \dot{m}_e$$

$$m_{\text{sys}} = w_A \cdot m_{\text{sys}}$$

$$\frac{d(w_A m_{\text{sys}})}{dt} = w_{Ai} \dot{m}_i - w_{Ae} \dot{m}_e$$

Same

$$\frac{dw_A}{dt} (m_{\text{sys}}) = (w_{Ai} - w_{Ae})(\dot{m}_i - \dot{m}_e)$$

$$\int_{w_0}^{w_f} \frac{1}{(w_{Ai} - w_{Ae})} dw_A = \int_{t_0}^{t_f} \frac{\dot{m}}{m_{\text{sys}}} dt = -\ln \left| \frac{w_{Ai} - w_{Ae}}{w_{Ai} - w_{Ae,0}} \right| = \frac{\dot{m}_{\text{sys}}}{m_{\text{sys}}} (t_f - t_0)$$

$$-\ln(w_{A\text{sys}}) =$$

$$t_f = \frac{m_{\text{sys}}}{\dot{m}} \left| \frac{w_{Ai}}{w_{Ai} - w_{Ae}} \right|$$

$$10 = \frac{10}{15} \left| \frac{0.1}{0.1 - w(t)} \right| \rightarrow 15 = \frac{1}{0.1 - w(t)} = 1.5 - 15w(t) = 0.1$$

$$w(t) = 0.093$$

r

$$\frac{15}{10} t_f = \frac{0.1}{0.1 - w(t)} \rightarrow w(t) =$$